

CO1 - DESCRIPTIVE WRITTEN
TEST :

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①. SMART - HOME AUTOMATION SYSTEMS :

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A smart home automation system is a intelligent agent which uses sensors, Artificial agent and actuators to perform its action, to control security, on and off algorithms, Home appliances. It receives inputs from motion, temperature, sensors and camera.

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Types of Agent :

- (1) Simple-Reflex agent
- (2) Model- Based agent
- (3) Goal- Based agent.

Simple-Reflex Agent :

A simple Reflex agents makes decisions only based on the current input using predefined conditions. Example: If motion is detected the lights are turned on.

ADVANTAGES:

Easy to implement, suitable for basic task, simple and can be used for many simple jobs.

DISADVANTAGES:

cannot remember previous states; cannot adapt quickly.

(2) MODEL - BASED AGENT:

It maintains internal model of the environment.

Example: System remember previous candidates of exam.

ADVANTAGES:

Works well with partially observable environment,
Makes better decisions.

DISADVANTAGES:

Requires more memory
More complex to design and
Analyse.

GOAL-BASED AGENT:

A Goal-Based agent is an agent that selects actions that help to achieve a specific goal.

It evaluates before making decisions.

Example:

Maintains the room temperature.

Ensures all lights are off ~~before~~ ~~no~~ at night.

ADVANTAGES:

Can plan future actions.

Flexible & Intelligent.

Achieves desired goal efficiently.

DISADVANTAGES:

Requires ~~Additional~~ computational

Decision-making takes more time.

PROBLEM STATEMENT :

A Robot in an Unknown maze doesn't know blueprint of the maze and doesn't know how to exit the maze. For it, the robot uses sensors find the obstacle and uses Uninformed cost search such as UCS, Uniform cost search and IDS (Iterative deepening search).

UNIFORM COST SEARCH (UCS) :

UCS expands the node having the low path cost first. It uses the cost of reaching a particular position as.

$$f(n) = g(n)$$

where $g(n)$ is the cost from starting node till the current node.

Example: If two paths are available UCS chooses the path with the lowest cost.

Therefore, it is useful when different paths in the maze has different path movement cost.

ADVANTAGES :

1. Complete when step costs are positive
2. Finds the least-cost solution
3. Suitable when different cost has different paths.

DISADVANTAGES :

1. Requires more memory
2. Can be slow.
3. May explore many unnecessary paths.

ITERATIVE DEEPENING SEARCH :

IDS repeatedly performs depth-limit search with increasing depth limit.

It first searches up to depth 0, then depth 1, then depth 2 continuous until goal is found.

ADVANTAGES :

- uses less memory
- completes for finite branching.
- Finds shortest solution when every step has the same cost.

Useful when depth of the solution is unknown.

DISADVANTAGES :

Repeat the same nodes during different iterations .

can take additional time .

Not suitable for different path cost.

TIME & SPACE COMPLEXITY :

$$\text{Time} = O(b^d).$$

$$\text{Space} = O(bd).$$

CONCLUSION :

The robot can be considered an intelligent agent because it senses the environment and make decisions, performs Actions.

For a normal maze, a model-based and goal-based agent with IDS is effective.